Case-disjoint benchmark of surrogate models for flow and thermal fields in supercritical CO_2 natural-circulation loops

Abstract

Surrogate models for supercritical carbon dioxide (sCO_2) natural-circulation loops (NCLs) can replace day-long steady-state conjugate computational fluid dynamics (CFD) simulations with millisecond-scale field predictions, yet most published evaluations rely on random node-level train–test splits or single pooled metrics that mask generalization failure across unseen operating conditions. Here we present a case-disjoint surrogate benchmark built from 42 legacy steady-state RANS simulations solved with a k–ε family turbulence model (k and ε are exported) and a Span–Wagner equation of state verified for internal thermodynamic consistency against `CoolProp` (median 0.0008%) across three heater–cooler configurations (`HHHC`, `HHVC`, and `VHVC`) and two loop scales. Across three random seeds, we benchmark 19 surrogate model variants (16 cross-configuration pooled models, including multiplicative `Domain-informed FiLM-MLP` and its matched input-concatenation ablation `Domain-informed Concat-MLP`, plus 3 within-group mesh-aligned models: `POD-RBF`, `POD-GPR`, and `Nearest case`) alongside a non-competing voxel discretization reference (`Lattice floor (oracle)`) under five-fold stratified case-disjoint cross-validation and leave-one-group-out structural/regime shift. We show that random node splitting understates five-fold case-disjoint error across the three strictly matched-configuration tree baselines (`Random Forest`, `Extra Trees`, `CatBoost`) by 6.78 × –8.25× on density, 11.29 × –30.08× on gauge pressure, and 5.62 × –10.24× on temperature in case-macro MAE (6.78 × –10.37× and 11.29 × –87.54× when including `XGBoost`) because high-capacity models memorize case-specific operating levels from adjacent mesh nodes. We also show that separating closed-loop gauge pressure into its case-level thermal-pressurization offset and centered spatial residual—benchmarked against a zero-residual baseline—prevents small within-case hydrostatic variations from distorting model rankings. Across the eight exported CFD fields, no single architecture wins universally: among cross-configuration pooled models, `DeepONet` and `PINN continuity` lead on density and temperature, `Extra Trees` best predicts case-level gauge pressure, and multiplicative feature-modulated coordinates (`Domain-informed FiLM-MLP`) achieve the lowest error across both turbulence fields (k, ε) and all three velocity components (u, v, w), while within-group `POD-RBF` yields the sharpest temperature and turbulence reconstructions when interpolating on a fixed loop mesh.

Keywords: supercritical CO_2, natural-circulation loop, CFD surrogate benchmark, case-disjoint validation, neural operators, proper orthogonal decomposition

1 Introduction

Natural-circulation loops (NCLs) operating with supercritical fluids remove heat without mechanical pumps by balancing buoyancy forces between heated and cooled sections against frictional and local losses [1, 2, 3]. Supercritical carbon dioxide (sCO_2, critical point $T_c = 304.13$ K $= 30.98°$C, $p_c = 7.3773$ MPa $= 73.773$ bar) is widely used as a scaled simulant fluid for supercritical water reactor (SCWR) thermal-hydraulics and compact power cycles because its lower critical point permits experiments and conjugate CFD studies near the pseudocritical locus [4, 5, 6]. Near the pseudocritical temperature, steep variations in density ρ, specific heat c_p, viscosity μ, and thermal conductivity k strongly couple buoyancy gradients with turbulence production and heat transfer [7, 8, 9].

Because resolving conjugate heat transfer between the primary sCO_2 pipe, the solid metallic wall, and the secondary cooling-water jacket requires fine near-wall meshes and real-gas property evaluations, each steady-state RANS simulation in our dataset requires roughly 24 hours of compute time. Data-driven spatial

surrogates—including gradient-boosted trees [10, 11, 12], coordinate neural networks [13, 14, 15], neural operators [16, 17, 18, 19], and proper orthogonal decomposition (POD) reduced-order models [20, 21]—provide millisecond-scale predictions once trained. However, three practical issues complicate surrogate benchmarking on closed natural-circulation loops: (i) random node-level train–test splits leak case-specific operating levels across neighboring mesh nodes; (ii) pooled metrics such as R^2 conflate between-case operating-level shifts with within-case spatial profile accuracy, while naive within-case normalization breaks down for gauge pressure in closed loops; and (iii) slender circular pipes ($D_i/L \sim 10^{-2}$) and varying heater–cooler orientations create different inductive-bias requirements for thermodynamic state fields versus velocity and turbulence fields.

To address these issues, this paper contributes: (1) a controlled comparison of 80/20 random-node splitting against five-fold case-disjoint cross-validation across three strictly matched-configuration tree ensembles, `XGBoost`, and two linear/constant controls; (2) a physical characterization and two-part evaluation protocol for closed-loop gauge pressure referenced against a zero-residual baseline; and (3) a three-seed benchmark of 19 surrogate model variants (16 cross-configuration pooled models and 3 within-group mesh-aligned models, plus an oracle voxel discretization floor) across eight CFD fields under both five-fold case-disjoint interpolation and leave-one-group-out structural/regime shift.

2 Physical Domain and sCO_2 Natural-Circulation Loop Dataset

2.1 Loop Geometry, CFD Solver Configuration, and Primary Fluid Domain

The dataset consists of 42 legacy converged steady-state RANS conjugate heat-transfer simulations of closed rectangular sCO_2 natural-circulation loops solved in ANSYS Fluent and exported on cross-sectional planes via ANSYS CFD-Post. All simulations use a k–ε family turbulence model (k and ε are exported as RANS closure variables rather than DNS/LES or experimental measurements) and a Span–Wagner equation of state verified for internal thermodynamic consistency against `CoolProp` [4, 22] (median relative density discrepancy 0.0008% between exported (T, p_{abs}) and ρ) for the primary sCO_2 fluid. Because the dataset is curated from archived CFD-Post plane exports, this `CoolProp` audit verifies equation-of-state consistency of the exported fields rather than mesh-independence or experimental flow validation.

The 42 cases cover three heater–cooler configurations—horizontal heater/horizontal cooler (`HHHC`), horizontal heater/vertical cooler (`HHVC`), and vertical heater/vertical cooler (`VHVC`)—across two loop scales (Table 1): 29 small-loop cases ($L_x \times L_y = 0.8936 \times 1.0000$ m, $D_i = 15.00$ mm) and 13 large-loop `VHVC` cases ($L_x \times L_y = 3.0100 \times 4.0800$ m, $D_i = 13.84$ mm). Complete radial dimensions of the stainless-steel wall and cooling-water jacket, axial heater/cooler spans, and sampling-plane counts (97 planes in small loops; 142 planes in the large loop) are given in Supplementary Section S1.

Each simulation is indexed by four boundary parameters encoded in its case name `{CONFIG}_P{Q}_{p}_bar_{mdot}kgs_T_{Tin}`: heater power $Q \in [10, 2500]$ W, nominal operating pressure $p_{\text{op}} \in \{80, 85, 90, 100\}$ bar, cooling-water mass flow rate $\dot{m} \in [0.010, 0.150]$ kg s^{-1}, and cooling-water inlet temperature $T_{\text{in}} \in \{283.5, 300, 305, 315\}$ K (Fig. 1). At each node (x, y, z), eight CFD target fields are exported: density ρ (kg m^{-3}), gauge pressure p_{gauge} (Pa), temperature T (°C), turbulence eddy dissipation ε (m^2 s^{-3}), turbulence kinetic energy k (m^2 s^{-2}), and velocity components (u, v, w) (m s^{-1}). Removing solid wall nodes (`null` velocity/turbulence, $\rho \approx 7970$ kg m^{-3}) and outer cooling-water jacket nodes ($\rho > 950$ kg m^{-3}) and deduplicating intersecting plane coordinates yields 5,451,341 unique primary sCO_2 nodes ($\rho \in [153.2, 892.5]$ kg m^{-3}; Fig. 2).

2.2 Behavior of Exported Gauge Pressure Across Cases

In all 42 simulations, the local absolute pressure used to evaluate CO_2 properties is the sum of the nominal operating pressure p_{op} and the exported gauge pressure $p_{\text{gauge}}(\mathbf{x})$:

$$p_{\text{abs}}(\mathbf{x}) = p_{\text{op}} + p_{\text{gauge}}(\mathbf{x}). \tag{1}$$

Table 1: Summary of the 42 steady-state CFD cases by heater–cooler configuration and loop scale. Full radial and plane-sampling details are in Supplementary Section S1.

Config.	Scale	Cases	$L_x \times L_y$ (m)	D_i (mm)	Q (W)	p_{op} (bar)	$\dot{m}$ (kg s^{-1})	sCO$_2$ nodes/case
HHHC	Small	20	0.894×1.000	15.00	1000–2000	80–100	0.050–0.150	97,264
HHVC	Small	5	0.894×1.000	15.00	10–50	90	0.050	89,771
VHVC	Small	4	0.894×1.000	15.00	10–40	90	0.050	89,647
VHVC	Large	13	3.010×4.080	13.84	250–2500	85	0.010–0.035	207,586
Total	**4 groups**	**42**	–	–	**10–2500**	**80–100**	**0.010–0.150**	**5,451,341 total**

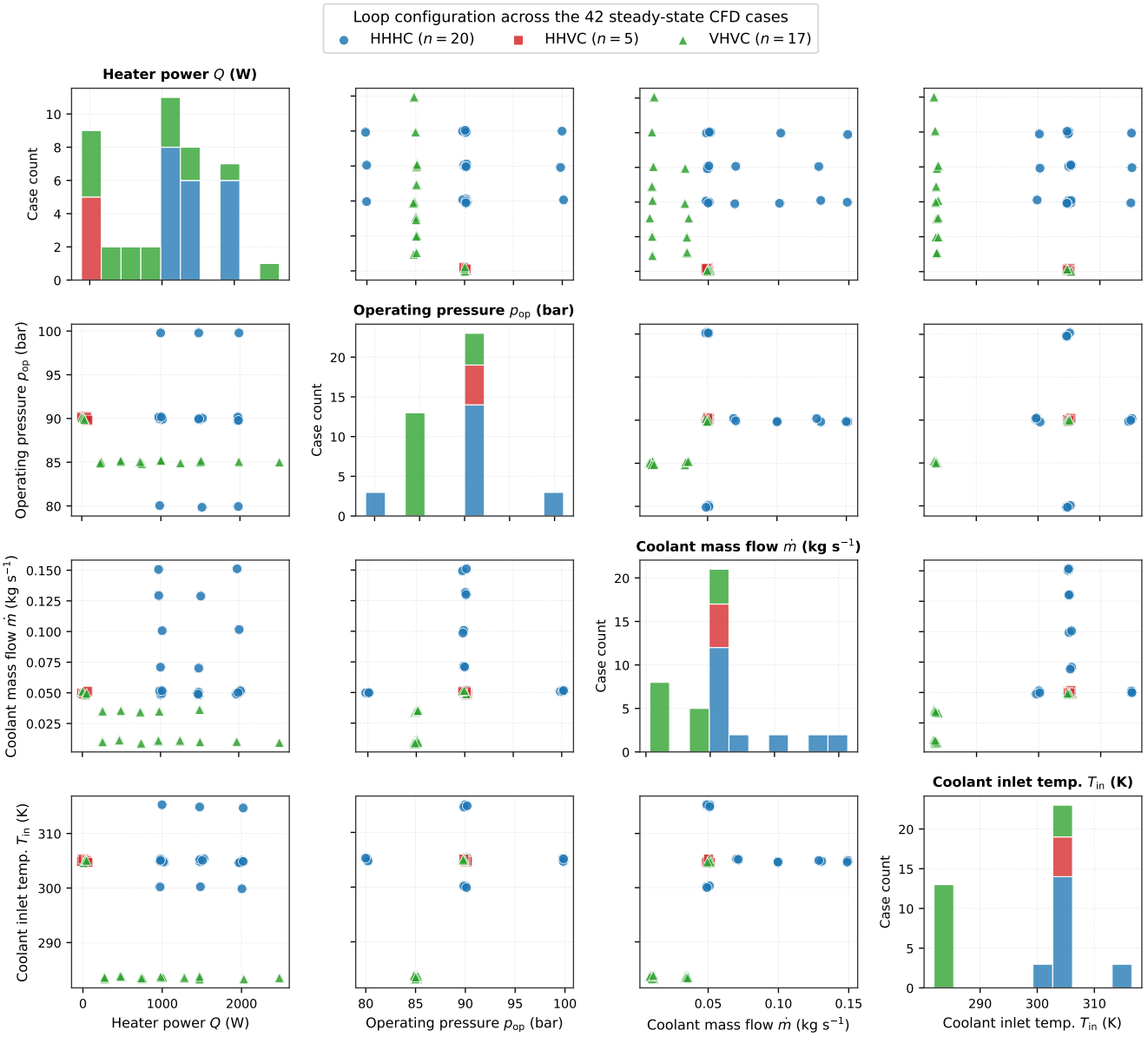


Figure 1: Operating-condition distribution across the 42 steady-state CFD cases by heater–cooler configuration (HHHC, HHVC, VHVC). Diagonal panels show stacked case-count histograms for Q, p_{op}, $\dot{m}$, and T_{in}.

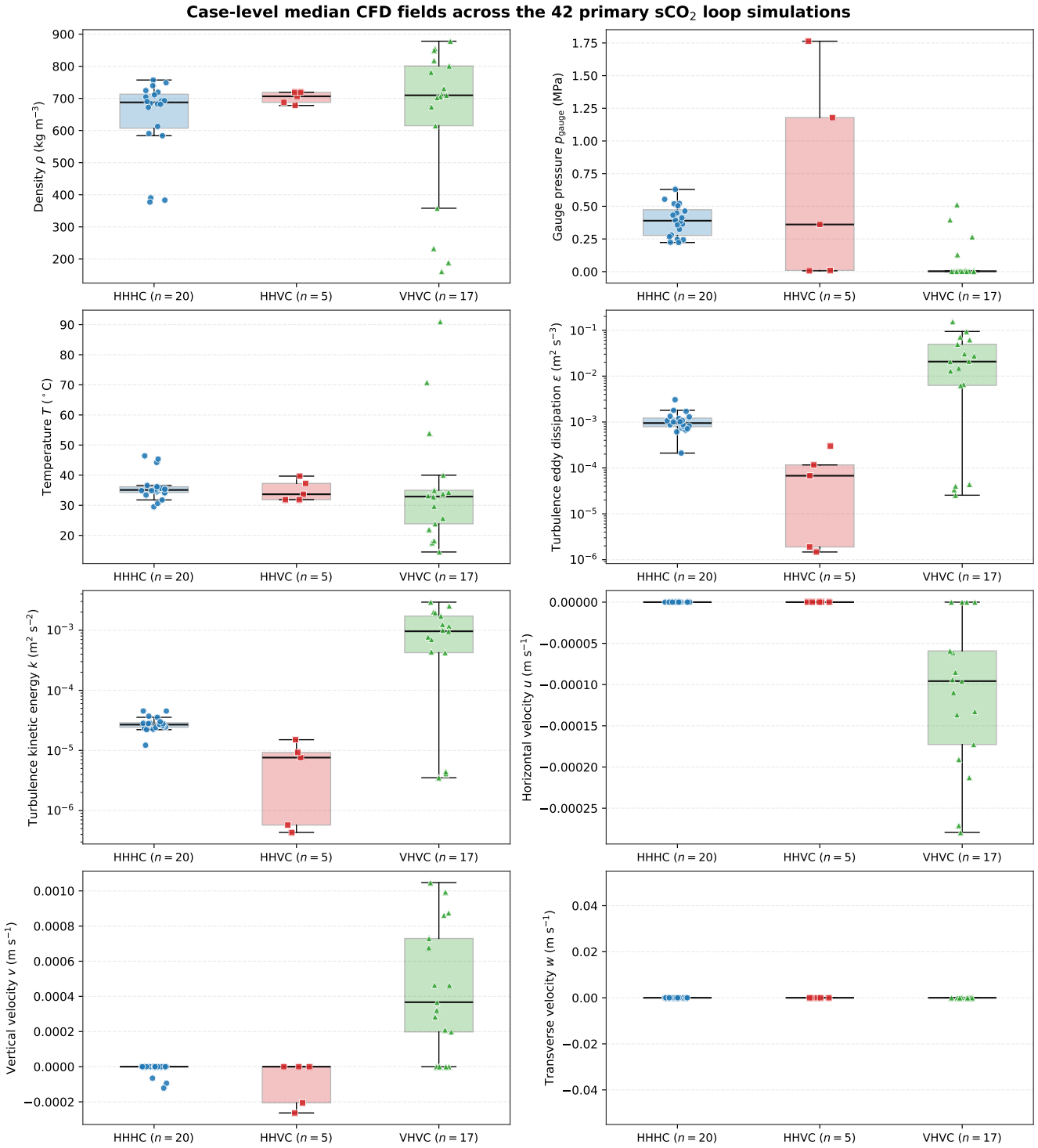


Figure 2: Case-level median distributions of the eight exported CFD fields across the 42 primary sCO_2 loop simulations (5, 451, 341 nodes), grouped by configuration (`HHHC`, `HHVC`, `VHVC`).

Evaluating $(T(\mathbf{x}), p_{\text{abs}}(\mathbf{x}))$ in `CoolProp` [4, 22] matches the exported CFD density $\rho(\mathbf{x})$ across all 42 cases to a median relative difference of 0.0008% (0.00004%–0.0015%). Within any single case c, p_{gauge} varies very little around the loop because the solver subtracts a spatially varying hydrostatic reference field $\rho_0 \mathbf{g} \cdot \mathbf{x}$ (subtracting a spatially constant offset would not reduce within-case variance): the within-case standard deviation $\sigma_c(p_{\text{gauge}})$ is only 0.15–0.39 kPa in the small loops and 1.45–8.79 kPa in the large loop (42-case mean $\bar{\sigma}_c = 1.51$ kPa).

Across different cases, however, the case-mean gauge pressure $\bar{p}_c$ spans 1.42 kPa to 1.763 MPa (Fig. 3):

- **Large loop vs. small loops:** In the 13 large-loop `VHVC` cases, the reference gauge pressure is pinned near 0 Pa at the top of the loop ($y = 4.08$ m), keeping $\bar{p}_c \in [1.42, 8.77]$ kPa (Fig. 3a). In the 29 small-loop cases (`HHHC`, `HHVC`, and small `VHVC`), the closed fluid domain has no top zero-gauge pin, so $\bar{p}_c$ rises strongly with the loop-mean temperature $\bar{T}_c$.

- **Empirical pressure–temperature slope vs. `CoolProp`:** Across the 17 small-loop cases at $p_{\mathrm{op}} = 90$ bar and $T_{\mathrm{in}} = 305$ K, a linear fit of $\bar{p}_c$ versus $\bar{T}_c$ gives an *empirical slope* of 0.214 MPa K^{-1} (0.235 MPa K^{-1}, or $\approx$ 0.23 MPa K^{-1}, across the five `HHVC` cases; Fig. 3b). By comparison, the theoretical constant-density derivative $(\partial p/\partial T)_\rho$ computed in `CoolProp` at each case's mean density and temperature is 0.315–0.365 MPa K^{-1} (median 0.338 MPa K^{-1}, or $\approx$ 0.35 MPa K^{-1} near $\rho = 710$ kg m^{-3} and $T = 305$–308 K; 0.133–0.417 MPa K^{-1} across all 29 small-loop cases in Supplementary Table S0b). Testing the fixed-mass hypothesis via the plane-weighted mean CO_2 density $\bar{\rho}_{\mathrm{plane},c}$ across cases at the same $(p_{\mathrm{op}}, T_{\mathrm{in}})$ shows that $\bar{\rho}_{\mathrm{plane},c}$ is not strictly constant: at 90 bar, 305 K it spans 671.6–704.0 kg m^{-3} (SD = 9.79 kg m^{-3}) in `HHHC`, 681.2–717.8 kg m^{-3} (SD = 16.66 kg m^{-3}) in `HHVC`, and 696.8–712.1 kg m^{-3} (SD = 6.62 kg m^{-3}) in small `VHVC` (and 161.5–860.8 kg m^{-3}, SD = 270.83 kg m^{-3} in top-pinned `VHVC/large`). Because fluid mass-charge and solver pressure-initialization settings were not separately recorded in the legacy CFD exports and $\bar{\rho}_{\mathrm{plane},c}$ decreases slightly in warmer cases ($\mathrm{d}\bar{\rho}/\mathrm{d}\bar{T} < 0$), the relation $\mathrm{d}\bar{p} \approx (\partial p/\partial T)_\rho \,\mathrm{d}\bar{T} + (\partial p/\partial \rho)_T \,\mathrm{d}\bar{\rho}$ yields an empirical slope lower than the constant-density `CoolProp` value.

- **Near-stagnant steady solutions in `HHVC` at 20 W and 30 W:** In the five `HHVC` cases at 90 bar and 305 K, the $Q = 20$ W and 30 W runs (`HHVC_P20_90_bar_0.05kgs_T_305` and `HHVC_P30_90_bar_0.05kgs_T_305`) are near-stagnant steady solutions (mean speed $|\overline{\mathbf{u}}| = 0.9$–$1.1$ mm s^{-1}, $\bar{T}_c = 31.92$–31.94°C, $\bar{p}_c = 6.1$–9.2 kPa), whereas the 10 W, 40 W, and 50 W runs have sustained buoyant circulation ($|\overline{\mathbf{u}}| = 0.044$–$0.107$ m s^{-1}, $\bar{T}_c = 33.58$–39.36°C, $\bar{p}_c = 0.361$–1.763 MPa; Fig. 3b).

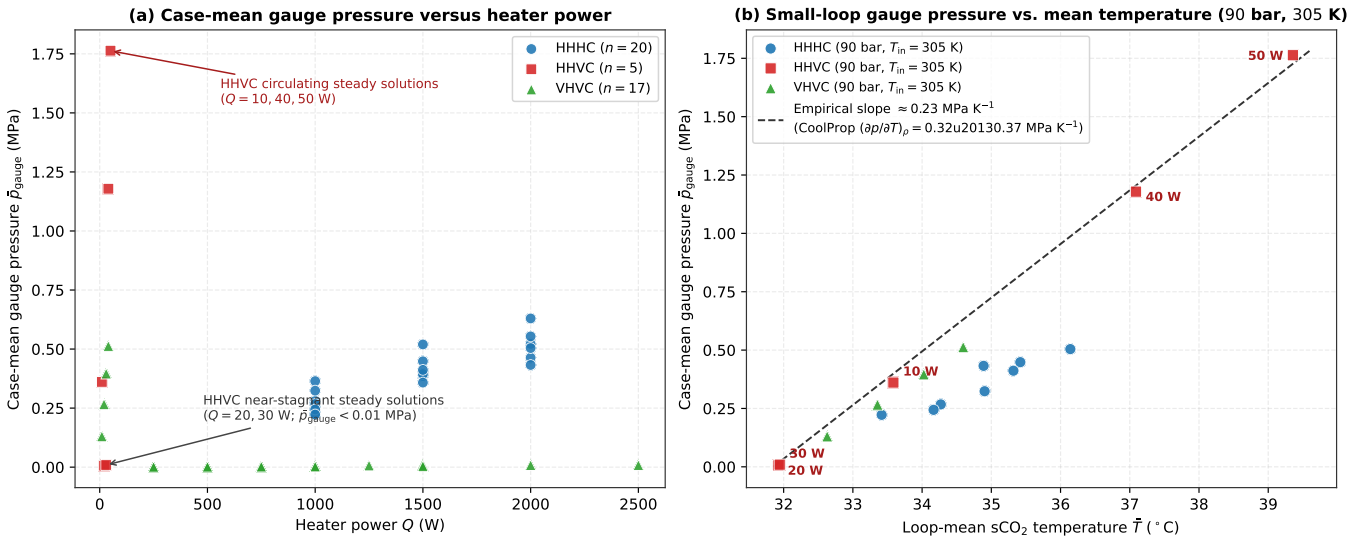


Figure 3: Case-level gauge-pressure behavior across the 42 sCO_2 simulations. (a) Case-mean gauge pressure $\bar{p}_{\mathrm{gauge}}$ versus heater power Q. (b) In the 17 small-loop cases at $p_{\mathrm{op}} = 90$ bar and $T_{\mathrm{in}} = 305$ K, $\bar{p}_{\mathrm{gauge}}$ follows an empirical slope of 0.214 MPa K^{-1} (0.235 MPa K^{-1} across `HHVC`) versus loop-mean temperature $\bar{T}_c$, compared against the `CoolProp` constant-density derivative $(\partial p/\partial T)_\rho = 0.315$–$0.365$ MPa K^{-1} (median 0.338 MPa K^{-1}).

3 Benchmark Methodology, Models, and Evaluation Protocol

3.1 Validation Protocols, Leakage Prevention, and Evaluation Metrics

Every model predicts the eight target fields $\mathbf{y}(\mathbf{x}, \mathbf{c}) \in \mathbb{R}^8$ at coordinates $\mathbf{x} = (x, y, z)$ given the case descriptor $\mathbf{c} = (Q, p_{\text{op}}, \dot{m}, T_{\text{in}}, L_x, L_y, D_i, \text{one-hot config})$. We evaluate three validation protocols:

1. **Five-fold stratified case-disjoint cross-validation:** For each seed $s \in \{2026, 2027, 2028\}$, the 42 cases are partitioned into five folds ($9, 9, 8, 8, 8$ test cases) stratified by the four groups (`HHHC/small`, `HHVC/small`, `VHVC/small`, `VHVC/large`). Each seed independently re-randomizes the five-fold case partition, the four validation cases per fold, the training node subsample ($30,000$ nodes/case in GPU 3-seed CV; $20,000$ in the single-seed CPU split test), and model initialization. Held-out test cases are evaluated on 100% of their mesh nodes ($5,451,341$ out-of-fold predictions per seed).

2. **Within-group mesh-aligned cross-validation:** Mesh-aligned models (`POD-RBF`, `POD-GPR`, `Nearest case`) require a shared mesh ordering and therefore train and predict strictly within the matching configuration/scale group of each five-fold split.

3. **Leave-one-group-out structural/regime shift:** All cases in one structural group (`HHHC/small`, $n = 20$; `HHVC/small`, $n = 5$; `VHVC/small`, $n = 4$; or `VHVC/large`, $n = 13$) are held out while the 16 cross-configuration pooled models train on the remaining three groups (16 models $\times$ 8 fields = 128 model–field evaluations per group). Because each group pairs a distinct heater–cooler arrangement or loop scale with its own operating-parameter range (Table 1), this protocol measures combined structural and operating-regime shift rather than pure geometric extrapolation at fixed operating conditions.

Verified train-only preprocessing and split hygiene: In the benchmark pipeline (`kaggle_clean_case_cv_predictions.ipynb`), four programmatic safeguards prevent data leakage across splits: (i) explicit set-disjointness assertions verify that training, validation, and test case ID sets are pairwise disjoint on every fold and group holdout before any preprocessing (`results/split_assignments.csv`); (ii) validation-case selection (`pick_val`) draws the four validation cases per fold exclusively from strata with at least eight non-test cases (`HHHC/small` and `VHVC/large`), ensuring that the small groups (`HHVC/small`, $n = 5$; `VHVC/small`, $n = 4$) retain all non-test cases (4 and 3–4 training cases per fold) for model fitting; (iii) input feature standardization (`cf_mean`, `cf_scale`) and target scaling (`TargetScaler`: $\operatorname{arcsinh}(y/\operatorname{median}(|y_{\text{train}}|))$ followed by z-score standardization on heavy-tailed ε and k, and z-score standardization on the other six fields) are fit strictly on the fold's training cases (`split["train_ids"]`); and (iv) for within-group mesh-aligned models (`run_group_method`), operating-parameter standardization, snapshot mean subtraction, truncated SVD basis construction (≤ 12 modes, 99.99% energy), and leave-one-out RBF hyperparameter selection (`tune_rbf`) are computed exclusively on the training cases belonging to that group ($\{c \in \mathcal{I}_{\text{train}} : g(c) = g\}$).

For each held-out case c (N_c nodes, truth $y_{c,i}$, prediction $\hat{y}_{c,i}$, case mean $\bar{y}_c = N_c^{-1} \sum_{i=1}^{N_c} y_{c,i}$, and population within-case standard deviation $\sigma_c = \sqrt{N_c^{-1} \sum_{i=1}^{N_c} (y_{c,i} - \bar{y}_c)^2}$ with `ddof=0`), we compute case-level MAE, RMSE, case-normalized MAE ($\text{nMAE}_c = \text{MAE}_c / \sigma_c$), and node-pooled R^2_{pooled}:

$$\text{MAE}_c = \frac{1}{N_c} \sum_{i=1}^{N_c} |\hat{y}_{c,i} - y_{c,i}|, \quad \text{RMSE}_c = \sqrt{\frac{1}{N_c} \sum_{i=1}^{N_c} (\hat{y}_{c,i} - y_{c,i})^2}, \quad R^2_{\text{pooled}} = 1 - \frac{\sum_{c,i} (\hat{y}_{c,i} - y_{c,i})^2}{\sum_{c,i} (y_{c,i} - \bar{y}_{\text{all}})^2}. \tag{2}$$

For each seed s, we aggregate across the 42 cases via the 42-case arithmetic mean for Macro MAE and Macro RMSE, the 42-case median and 90th percentile (P90) for nMAE_c, and pooled R^2_{pooled} across all $5,451,341$ nodes, and then report the mean $\pm$ SD across the three seeds. Because $\bar{\sigma}_c(p_{\text{gauge}}) = 1.51$ kPa is small relative to between-case pressure shifts (Section 2.2), we suppress raw nMAE_c for p_{gauge} and decompose each case's

pressure error into its case-mean level error $e_{\text{level},c}$ and centered spatial residual $\tilde{p}_{c,i} = p_{c,i} - \bar{p}_c$ (satisfying $\text{RMSE}_c^2 = (\hat{\bar{p}}_c - \bar{p}_c)^2 + \text{RMSE}_{\text{res},c}^2$ identically under `ddof=0`):

$$e_{\text{level},c} = |\hat{\bar{p}}_c - \bar{p}_c|, \quad \text{RMSE}_{\text{res},c} = \sqrt{\frac{1}{N_c}\sum_{i=1}^{N_c}\left(\hat{\tilde{p}}_{c,i} - \tilde{p}_{c,i}\right)^2}, \quad \text{nRMSE}_{\text{res},c} = \frac{\text{RMSE}_{\text{res},c}}{\max(\sigma_c, 1\ \text{kPa})}. \tag{3}$$

For each seed, $e_{\text{level},c}$ is averaged across the 42 cases, $\text{RMSE}_{\text{res},c}$ is aggregated via the 42-case root-mean-square $\text{RMSE}_{\text{res}} = \sqrt{\frac{1}{42}\sum_{c=1}^{42}\text{RMSE}_{\text{res},c}^2}$, and $\text{nRMSE}_{\text{res},c}$ is summarized by its 42-case median before averaging across the three seeds. Predicting a spatially constant pressure within each case ($\hat{\tilde{p}}_{c,i} = 0$, achieved by `Mean baseline`, where $\text{RMSE}_{\text{res},c} = \sigma_c$) defines the **zero-residual baseline**, which yields a three-seed mean RMSE_{res} of 2.89 kPa (2886 Pa; 42-case arithmetic mean 1.51 kPa) and median $\text{nRMSE}_{\text{res}}$ of 0.202.

3.2 Benchmark Model Suite, Optimization Protocol, and Verified Hyperparameters

Table 2 summarizes all 19 evaluated surrogate model variants (16 cross-configuration pooled models and 3 within-group aligned models) plus the non-competing `Lattice floor (oracle)` discretization reference, together with parameter counts, per-fold training times, and verified hyperparameters from the GPU and CPU benchmark scripts. All pointwise neural networks train with case-balanced mini-batches of 8,192 nodes per step (drawing an equal number of nodes from each training case in the fold), cosine learning-rate decay, gradient clipping at 1.0, and early stopping (patience 10, evaluated every 200 steps on the four held-out validation cases per fold, capped at 2600 steps or 8 min/fold on an NVIDIA P100 GPU). To isolate multiplicative feature modulation in coordinate networks, `Domain-informed FiLM-MLP` (1,069,064 parameters) modulates each hidden block by $\mathbf{h}^{(\ell)} \odot (1 + \boldsymbol{\gamma}^{(\ell)}(\mathbf{c})) + \boldsymbol{\beta}^{(\ell)}(\mathbf{c})$ [15], whereas its matched ablation `Domain-informed Concat-MLP` (344,072 parameters, lacking the per-block linear FiLM generators) concatenates $\mathbf{c}$ at the input layer. In `PINN continuity`, the steady mass-conservation residual is computed in physical units via automatic differentiation with respect to normalized coordinates $\tilde{x}_a \in [0,1]$ divided by each case's physical bounding-box span $\Delta x_{c,a}$: $\mathcal{R}_{\text{cont}} = S_{\text{flux}}^{-1}\sum_{a\in\{x,y,z\}}(\Delta x_{c,a})^{-1}\partial(\hat{\rho}\hat{u}_a)/\partial\tilde{x}_a$, where $S_{\text{flux}} = (|\mu_\rho| + \sigma_\rho)\bar{\sigma}_u/\overline{\Delta x}$ normalizes the divergence residual before multiplying by $\lambda_{\text{cont}} = 0.05$.

4 Results and Discussion

4.1 Optimism of Node-Random Splits Versus Case-Disjoint Validation

Table 3 and Figure 4 compare 80/20 random node splitting against five-fold case-disjoint cross-validation on the 5,451,341 primary sCO_2 nodes from the paired single-seed (`2026`) split-comparison benchmark (`results/co2_candidate_case_and_random_node_metrics.csv`). Whereas the two low-capacity controls (`Mean baseline` and `Ridge`) show error ratios near 1.0× (0.95×–1.26×) because they cannot memorize local mesh neighborhoods, the three strictly matched-configuration tree models (`Random Forest`, `Extra Trees`, `CatBoost`, which use identical tree counts and depths in both protocols) suffer large error inflation under case-disjoint evaluation on thermodynamic state fields: 6.78×–8.25× on density, 11.29×–30.08× on gauge pressure, and 5.62×–10.24× on temperature in case-macro MAE (3.52×–4.05× and 2.95×–4.39× in median case nMAE on ρ and T). Including `XGBoost` (which used `max_depth=6` on random-node splits vs. `max_depth=8` on case-disjoint splits) widens the four-tree case-macro MAE inflation range to 6.78×–10.37× on ρ and 11.29×–87.54× on p_{gauge}. This inflation occurs because random node splits allow high-capacity trees to interpolate case-specific operating levels from adjacent mesh nodes of the same simulation.

4.2 Three-Seed Case-Disjoint Benchmark Across All Eight CFD Fields

Figure 5 shows the three-seed (`2026`, `2027`, `2028`) five-fold case-macro MAE of 18 models relative to the five-fold `Mean baseline` (omitting `POD-GPR` because its error profile closely tracks `POD-RBF`), and Table 4 reports the detailed five-fold metrics—including the 90th-percentile (P90) case nMAE and the non-competing

Table 2: Evaluated surrogate models (16 pooled + 3 within-group aligned + 1 oracle discretization floor), parameter counts, mean GPU five-fold training time per fold, and verified hyperparameters.

Model	Family (time/fold)	Conditioning representation, parameter count, & verified hyperparameters
`Mean baseline`	Constant (4.5 s)	Predicts training-fold node mean per target field (zero-residual baseline $\tilde{p}_{c,i} = 0$ on p_{gauge}).
`Ridge`	Linear (2.2 s)	ℓ_2 linear regression ($\alpha = 10.0$) on standardized $[\mathbf{x}, \mathbf{c}]$.
`Random Forest`	Bagged (35.5 s)	Multi-output CART on $[\mathbf{x}, \mathbf{c}]$: 48 trees (80 in CPU split test; matched across both splits), max depth 20, max leaf nodes 4096 (2048 CPU), min leaf 5, max features 0.8, bootstrap 0.3.
`Extra Trees`	Random (20.2 s)	Multi-output randomized trees on $[\mathbf{x}, \mathbf{c}]$: 64 trees (96 CPU; matched across splits), depth 20, max leaf 4096.
`XGBoost`	Boosted (29.5 s)	Per-target `hist` trees: 300 trees (220 CPU), depth 8 (depth 6 on random-node CPU), lr 0.08, subsample 0.85, colsample 0.85, $\lambda = 2.0$.
`CatBoost`	Oblivious (19.3 s)	Multi-output symmetric trees (`MultiRMSE`): 350 iterations, depth 8, lr 0.08, $\ell_2 = 3.0$ (matched across splits).
`DeepONet`	Operator (184 s)	$699,528$ params; 4-layer branch MLP on $\mathbf{c}$ + 4-layer trunk MLP on $\mathbf{x}$ (width 256, latent 128), MSE loss, AdamW (lr $= 5 \times 10^{-4}$), case-balanced batch 8192.
`PINN data-only`	Coord. (86.8 s)	$277,512$ params; 5-layer Fourier-coordinate MLP ($B = 4$ bands, width 256, GELU + LayerNorm), MSE loss.
`PINN continuity`	Physics (235 s)	$277,512$ params; identical to `PINN data-only` + $\lambda_{\text{cont}} = 0.05\,\overline{\mathcal{R}^2_{\text{cont}}}$ on 1024 collocation nodes/step.
`Domain-informed FiLM-MLP`	FiLM (384 s)	$1,069,064$ params; 5-block residual MLP (width 256, $B = 4$ Fourier bands + leg centerline/arc-length features), per-block FiLM modulation by $\mathbf{c}$, Huber loss ($\delta = 1.0$), AdamW (lr $= 5 \times 10^{-4}$, batch 8192).
`Domain-informed Concat-MLP`	Ablation (111 s)	$344,072$ params; identical trunk and training to `FiLM-MLP`, but concatenates $\mathbf{c}$ at the input layer without FiLM generators.
`Latent Neural Operator`	Latent (666 s)	$583,560$ params; cross-attention from Fourier coordinates ($B = 4$) to $M = 24$ latent tokens (width 128, 4 heads, depth 3/2), Huber loss.
`Transolver-style`	Slice (430 s)	$715,784$ params; 4-block physics-attention network (width 128, 4 heads, $S = 32$ slices), 8192 context / 4096 query nodes/step, AdamW (lr $= 3 \times 10^{-4}$, 2200 steps).
`GAMS-GNN-style`	GNN (746 s)	$869,096$ params; 4-block message passing (width 96) across $k = 6$ fine kNN graph (8192 nodes) and $(20, 20, 6)$ / $(10, 10, 3)$ voxel graphs, AdamW (lr $= 2 \times 10^{-4}$, 1600 steps).
`F-FNO`	FNO (138 s)	$155,888$ params; 4 factorized 1D spectral blocks (width 24, 12 modes) on $64 \times 64 \times 16$ voxel grid, AdamW (lr $= 10^{-3}$, 2000 steps).
`U-FNO3D` (`U-FNO`)	U-FNO (125 s)	$212,408$ params; 3 spectral blocks + 3D U-Net branch (width 24, 12 modes) on $64 \times 64 \times 16$ voxel grid.
`POD-RBF` / `POD-GPR`	ROM (2.4 s)	Within-group train-only SVD (≤ 12 modes, 99.99% energy) + Gaussian RBF (train-only LOO-CV over $\ell \in [0.25, 8], \alpha \in [10^{-6}, 1]$) or GPR ($\ell = 1, \alpha = 10^{-4}$).
`Nearest case`	Lookup (1.8 s)	Copies the closest training case in standardized operating-parameter space within the same group.
`Lattice floor (oracle)`	Floor (1.1 s)	Non-competing discretization floor: splats true test-case CFD values onto the $64 \times 64 \times 16$ grid and trilinearly interpolates back to mesh nodes.

`Lattice floor (oracle)` voxelization reference—for the leading pooled and within-group models on each field (complete 20-row tables for all 19 competing models plus `Lattice floor (oracle)` are in Supplementary Tables S1–S8). Every ratio Macro MAE/Mean-base MAE in Table 4 matches the corresponding cell of Figure 5 to the displayed precision.

Four main findings emerge from Table 4 and Figure 5:

Table 3: Error inflation ratios (five-fold case-disjoint error divided by 80/20 random-node error, seed 2026) separating the three strictly matched-configuration tree models (`Random Forest`, `Extra Trees`, `CatBoost`) from `XGBoost` (depth 8 vs. 6) and the low-capacity `Mean baseline` and `Ridge` controls.

Target field	**3 matched trees (Macro MAE ratio)**	**3 matched trees (Median nMAE ratio)**	`XGBoost` **(MAE / nMAE)**	`Mean` control (MAE / nMAE)	`Ridge` control (MAE / nMAE)
Density ρ	**6.78–8.25**×	**3.52–4.05**×	**10.37**× / **3.34**×	0.98× / 0.90×	1.15× / 1.05×
Gauge pressure p_{gauge}	**11.29–30.08**×	n/a[†]	**87.54**× / n/a[†]	1.03× / n/a	1.26× / n/a
Temperature T	**5.62–10.24**×	**2.95–4.39**×	**7.43**× / **3.17**×	0.95× / 0.72×	1.06× / 1.02×
TED ε	1.02–1.46×	1.01–1.09×	1.29× / 0.91×	0.97× / 0.87×	1.01× / 1.00×
TKE k	1.39–2.99×	1.05–1.22×	2.09× / 0.89×	0.98× / 0.87×	1.10× / 1.00×
Velocity u	1.77–2.76×	1.62–2.03×	2.04× / 1.15×	0.99× / 0.99×	1.04× / 1.05×
Velocity v	1.59–2.23×	1.32–1.74×	1.68× / 0.95×	0.99× / 1.01×	1.03× / 0.99×
Velocity w	1.00–1.02×	1.00–1.02×	0.95× / 0.90×	1.00× / 1.01×	1.00× / 1.02×

[†] Within-case nMAE is suppressed for gauge pressure because $\bar{\sigma}_c = 1.51$ kPa reflects only the within-case vertical pressure variation.

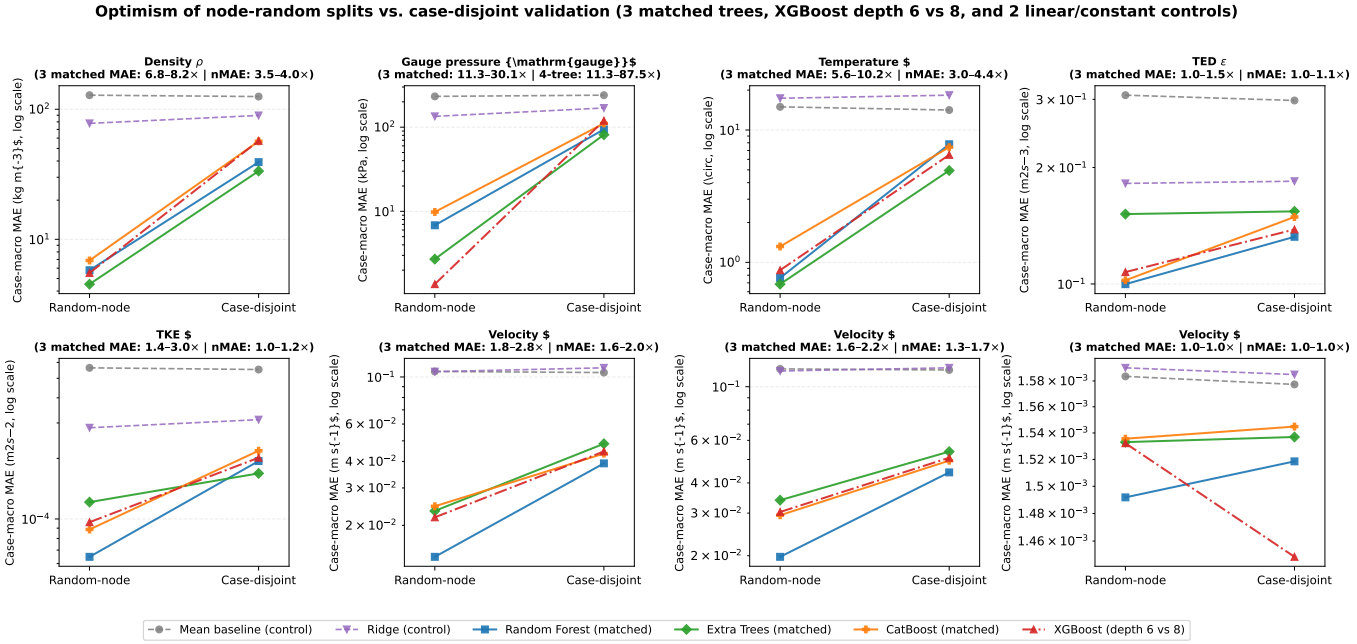


Figure 4: Case-macro MAE (log scale) under 80/20 random-node splits versus five-fold case-disjoint validation across the three strictly matched tree baselines (`RF`, `ET`, `CB`), `XGBoost` (depth 6 vs. 8), and two controls (`Mean`, `Ridge`).

1. **State fields (ρ, T) vs. kinematic/turbulence fields $(u, v, w, k, \varepsilon)$:** On density and temperature, pooled models reduce case-macro MAE by 72.8%–79.7% (ρ) and 63.2% (T) and reach high pooled R^2 (0.862–0.936 and 0.736), but narrow within-case standard deviations in the small loops ($\sigma_c(T) \approx$ 0.25–2.5°C) keep median case nMAE at 0.837 (`DeepONet` on ρ, P90 5.60) and 1.350 (`PINN continuity` on T, P90 6.11; reduced to median 0.974 and P90 4.33 by within-group `POD-RBF`). Across structural groups, `Domain-informed FiLM-MLP` achieves low median nMAE on the large-scale `VHVC/large` (0.471 on ρ, 0.482 on T, 0.082 on u, 0.088 on v) and `VHVC/small` (0.251 on ρ, 0.500 on T) groups, whereas `HHHC/small` (ρ: 1.391, T: 2.458) and `HHVC/small` (ρ: 2.273, T: 5.363, which contains the two near-stagnant 20–30 W cases) drive the upper-tail P90 errors. On velocity and turbulence fields, where case-mean offsets are near zero and spatial profiles are geometry-locked, `Domain-informed FiLM-MLP` achieves median case nMAE of 0.162 (u), 0.175 (v), 0.192 (ε), and 0.242 (k).

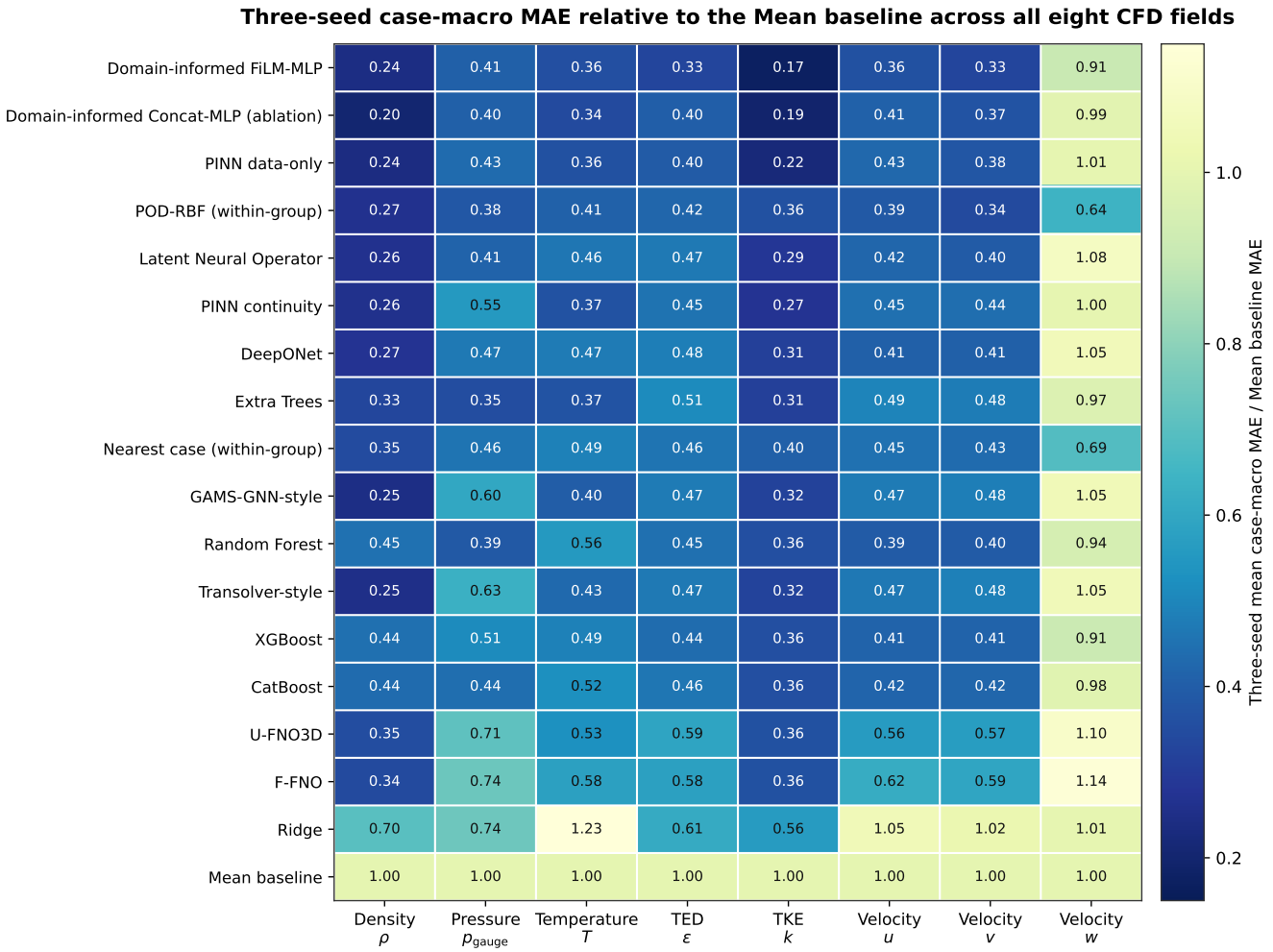


Figure 5: Three-seed mean five-fold case-macro MAE relative to the five-fold `Mean baseline` across all eight CFD fields, ordered by geometric-mean relative error (18 models shown; full 20-row tables including `POD-GPR` and `Lattice floor (oracle)` are in Supplementary Tables S1–S8).

2. **FiLM modulation vs. input concatenation:** Relative to `Domain-informed Concat-MLP` (344k params), multiplicative FiLM conditioning (1.07M params) reduces median case nMAE on six of the seven spatially varying fields: T (1.467 vs. 1.758), ε (0.192 vs. 0.226), k (0.242 vs. 0.273), u (0.162 vs. 0.224), v (0.175 vs. 0.230), and w (0.272 vs. 0.318), whereas concatenation yields lower case-macro MAE on density (26.3 vs. 31.3 kg m^{-3}, P90 nMAE 3.74 vs. 5.66).

3. **Gauge-pressure level vs. spatial residual and the zero-residual baseline:** Total gauge-pressure MAE (83.2 kPa for `Extra Trees`, a 64.7% reduction below the `Mean baseline` of 235.9 kPa) is dominated by the case-level offset $e_{\mathrm{level},c}$ (82.9 kPa). On the centered spatial residual $\tilde{p}_{c,i}$ (Table 4 and Supplementary Table S8), the zero-residual baseline ($\hat{\tilde{p}}_{c,i} = 0$, achieved by `Mean baseline`) already attains $\mathrm{RMSE}_{\mathrm{res}} = 2.89$ kPa (2886 Pa) and $\mathrm{nRMSE}_{\mathrm{res}} = 0.202$. Only mesh-aligned `Nearest case` ($\mathrm{RMSE}_{\mathrm{res}} = 0.67$ kPa, $\mathrm{nRMSE}_{\mathrm{res}} = 0.0122$), `POD-GPR` (0.46 kPa, 0.0457), `POD-RBF` (0.47 kPa, 0.0483), and `XGBoost` (1.01 kPa, 0.151) improve upon predicting zero spatial residual, whereas coordinate networks and neural operators inject spurious spatial fluctuations ($\mathrm{RMSE}_{\mathrm{res}} = 24.4$–$189.6$ kPa, $\mathrm{nRMSE}_{\mathrm{res}} = 2.91$–$24.3$).

4. **Oracle voxelization error explains FNO underperformance on slender pipe loops:** Evaluating `Lattice floor (oracle)` (Table 4) shows that simply projecting the ground-truth CFD fields onto the $64 \times 64 \times 16$ Cartesian grid and trilinearly interpolating back—with zero learning error—

Table 4: Three-seed (2026, 2027, 2028) five-fold out-of-fold accuracy across the 42 sCO_2 cases. For each target, the leading cross-configuration pooled model is listed first, followed by the best within-group mesh-aligned model where it achieves lower error, along with the non-competing `Lattice floor (oracle)` $64 \times 64 \times 16$ voxel discretization floor.

Target (unit)	Model	Scope	Macro MAE	Mean-base MAE	Reduction (%)	Macro RMSE	Median nMAE (±SD)	P90 nMAE	Pooled R^2
ρ (kg m^{-3})	`DeepONet`	Pooled	35.1	129.3	72.8%	37.1	0.837 ± 0.160	5.60	0.862
	`Concat-MLP`	Pooled	26.3	129.3	79.7%	29.1	1.011 ± 0.069	3.74	0.936
	`Lattice floor`	Oracle	4.78	129.3	96.3%	8.15	0.336 ± 0.000	0.461	0.996
p_{gauge} (kPa)	`Extra Trees`	Pooled	83.2	235.9	64.7%	83.7	2.91†	6.99‡	0.815
	`POD-RBF`	Aligned	90.1	235.9	61.8%	90.2	0.0483†	0.47‡	0.571
	`Nearest case`	Aligned	108.1	235.9	54.2%	108.2	0.0122†	0.67‡	0.776
	`Mean baseline`	Zero-res	235.9	235.9	0.0%	235.9	0.202†	2.89‡	−0.009
	`Lattice floor`	Oracle	0.0094	235.9	100.0%	0.0141	0.0018†	0.0268‡	1.000
T (°C)	`PINN continuity`	Pooled	5.48	14.90	63.2%	7.32	1.350 ± 0.044	6.11	0.736
	`POD-RBF`	Aligned	6.11	14.90	59.0%	7.42	0.974 ± 0.041	4.33	0.620
	`Lattice floor`	Oracle	0.768	14.90	94.8%	1.24	0.334 ± 0.000	0.452	0.992
ε (m^2 s^{-3})	`FiLM-MLP`	Pooled	0.0978	0.3002	67.4%	0.315	0.192 ± 0.020	0.474	0.569
	`POD-RBF`	Aligned	0.127	0.3002	57.8%	0.369	0.177 ± 0.018	1.46	−0.575
	`Lattice floor`	Oracle	0.171	0.3002	43.2%	0.409	0.339 ± 0.000	0.403	0.072
k (10^{-4} m^2 s^{-2})	`FiLM-MLP`	Pooled	0.947	5.547	82.9%	1.57	0.242 ± 0.051	0.679	0.898
	`POD-RBF`	Aligned	2.00	5.547	64.0%	2.78	0.237 ± 0.015	1.25	−0.070
	`Lattice floor`	Oracle	2.34	5.547	57.8%	3.74	0.415 ± 0.000	0.446	0.464
u (m s^{-1})	`FiLM-MLP`	Pooled	0.0381	0.1051	63.7%	0.0569	0.162 ± 0.015	0.810	0.801
	`Lattice floor`	Oracle	0.0355	0.1051	66.2%	0.0628	0.196 ± 0.000	0.265	0.779
v (m s^{-1})	`FiLM-MLP`	Pooled	0.0383	0.1175	67.4%	0.0591	0.175 ± 0.004	0.706	0.861
	`Lattice floor`	Oracle	0.0439	0.1175	62.6%	0.0749	0.241 ± 0.000	0.344	0.793
w (10^{-3} m s^{-1})	`FiLM-MLP`	Pooled	1.43	1.577	9.3%	3.46	0.272 ± 0.007	0.486	0.418
	`POD-RBF`	Aligned	1.01	1.577	35.9%	2.55	0.207 ± 0.008	0.378	0.650
	`Nearest case`	Aligned	1.08	1.577	31.4%	2.93	0.185 ± 0.009	0.412	0.551
	`Lattice floor`	Oracle	1.50	1.577	4.9%	4.02	0.323 ± 0.000	0.348	0.159

† For gauge pressure, the Median nMAE column reports the centered spatial residual $\text{nRMSE}_{\text{res}}$ (1 kPa denominator floor; Table S8). ‡ For gauge pressure, the P90 column reports the physical-unit centered spatial residual RMSE_{res} in kPa (where `Mean baseline` represents the zero-residual prediction $\tilde{p}_{c,i} = 0$ with $\text{RMSE}_{\text{res}} = 2.89$ kPa and $\text{nRMSE}_{\text{res}} = 0.202$).

already incurs median case nMAE of 0.336 (ρ), 0.334 (T), 0.339 (ε), 0.415 (k), 0.196 (u), 0.241 (v), and 0.323 (w). Because the voxel width in the large loop (47.0 × 63.8 mm) is 3.4 × –4.6× wider than the pipe diameter (D_i = 13.84 mm), this discretization floor alone exceeds the out-of-fold error of `Domain-informed FiLM-MLP` on every velocity and turbulence field, explaining why `F-FNO` and `U-FNO3D` rank near the bottom of the neural models (Fig. 5).

Exploratory statistical comparisons and paired case-level effect sizes: Treating nonparametric Friedman tests and Holm-adjusted two-sided Wilcoxon signed-rank tests across the 42 cases (computed on three-seed case-averaged five-fold nMAE, accompanied by the median paired difference $\Delta = \text{Median}_c(\text{nMAE}_{\text{comp},c} - \text{nMAE}_{\text{FiLM},c})$ and its 95% bootstrap confidence interval; Supplementary Table S10) as exploratory, overall model differences are significant on all eight fields ($p < 0.001$). Against `Domain-informed FiLM-MLP` ($\alpha = 0.05$): (i) on ε and k, `FiLM-MLP` ranks first (mean rank 4.62), significantly outperforming

all 15 pooled baselines on ε ($\Delta = +0.034$ $[+0.021, +0.071]$ vs. `Concat-MLP`, $+0.073$ $[+0.056, +0.098]$ vs. `DeepONet`, $+0.124$ $[+0.087, +0.141]$ vs. `Extra Trees`) and 14 of 15 on k ($\Delta = +0.150$ $[+0.077, +0.176]$ vs. `DeepONet`, $+0.186$ $[+0.142, +0.230]$ vs. `Extra Trees`; all except `Concat-MLP`, $\Delta = +0.034$ $[-0.002, +0.057]$, $p_{\text{Holm}} = 0.190$), while not differing significantly from within-group `POD-RBF` ($\Delta = -0.023$ on ε, $+0.012$ on k), `POD-GPR`, and `Nearest case` ($p_{\text{Holm}} \geq 0.190$); (ii) on u and v, `FiLM-MLP` ranks first (5.83 and 5.26), significantly outperforming `Concat-MLP` ($\Delta = +0.029$ $[+0.017, +0.057]$ on u, $+0.035$ $[+0.021, +0.066]$ on v), both `PINNs` ($\Delta = +0.066$ and $+0.100$), `LNO`, `Transolver-style`, `GAMS-GNN-style`, both `FNOs` (`U-FNO3D`: $\Delta = +0.153$ and $+0.184$), `Ridge`, and `Mean baseline` ($p_{\text{Holm}} < 0.05$, plus `DeepONet` $\Delta = +0.086$ and `Extra Trees` $\Delta = +0.103$ on v), while not differing significantly from the remaining tree ensembles or within-group aligned models ($p_{\text{Holm}} \geq 0.052$); (iii) on w, `FiLM-MLP` is significantly outperformed by within-group `POD-RBF` ($\Delta = -0.097$ $[-0.136, -0.069]$), `POD-GPR`, and `Nearest case`, does not differ significantly from `PINN continuity`, tree ensembles, `Ridge`, or `Mean baseline` ($p_{\text{Holm}} \geq 0.096$), and significantly outperforms all other neural/operator baselines ($\Delta = +0.041$ $[+0.026, +0.051]$ vs. `Concat-MLP`, $p_{\text{Holm}} < 0.05$); and (iv) on ρ and T, `FiLM-MLP` is significantly outperformed by `POD-RBF` on T ($\Delta = -0.708$ $[-1.382, -0.048]$), does not differ significantly from `DeepONet` ($\Delta = +0.076$ on ρ, $+0.151$ on T), `Concat-MLP` ($\Delta = -0.060$ on ρ, $+0.094$ on T), `PINNs`, `Transolver-style`, `GAMS-GNN-style`, or `Extra Trees` ($p_{\text{Holm}} \geq 0.170$), and significantly outperforms `F-FNO`, `U-FNO3D`, `Ridge`, and `Mean baseline`.

4.3 Spatial Reconstruction Analysis and Leave-One-Group-Out Shift

Figure 6 shows held-out CFD fields, surrogate predictions, and absolute error maps (seed `2026`, labeled by nMAE_c rank) for showcase cases on ρ (`DeepONet`, rank 15/42; 3-seed rank 3/42), T (`PINN continuity`, rank 17/42; 3-seed rank 1/42), and v (`FiLM-MLP`, rank 10/42; 3-seed rank 3/42), while Figure 7 directly presents the corresponding **median (rank 21/42)** and **worst (rank 42/42)** held-out cases per field in the main text. The median (rank 21/42) cases capture loop-scale buoyancy stratification and vertical core velocity gradients with errors concentrated along the cooling-jacket boundary layer and pipe elbows, whereas the worst (rank 42/42) cases occur at an unrepeated extreme boundary condition (`HHHC_P2000_90_bar_0.05kgs_T_315` on ρ and T, the sole simulation at $T_{\text{in}} = 315$ K with $\bar{T}_c = 47.5$°C) or in the near-stagnant `HHVC_P20_90_bar_0.05kgs_T_305` run on v ($\sigma_c(v) = 1.5$ mm s^{-1}).

Table 5 summarizes the leave-one-group-out shift study across the four structural/regime groups (16 pooled models × 8 fields = 128 model–field evaluations per group; full 16-model metrics in Supplementary Table S9). To avoid post-hoc selection bias from reporting only the retrospective winner per cell, Table 5 reports a **fixed preselected shortlist of seven models** (`Mean baseline`, `Extra Trees`, `XGBoost`, `DeepONet`, `PINN continuity`, `Domain-informed Concat-MLP`, and `Domain-informed FiLM-MLP`) alongside the best observed pooled model across all 16 candidates. When `VHVC/small` ($n = 4$) is held out, both the small-loop geometry (0.894×1.000 m, 90 bar, 305 K) and the `VHVC` orientation are represented in the training set, enabling `XGBoost`, `Extra Trees`, `DeepONet`, and `FiLM-MLP` to improve substantially upon the `Mean baseline` on ρ, T, ε, and k. Conversely, holding out `VHVC/large` ($n = 13$, which shifts loop dimensions by $3.4\times$ –$4.1\times$ and T_{in} to 283.5 K) or `HHHC/small` ($n = 20$, which removes all horizontal-cooler cases at $Q = 1000$–2000 W) degrades all surrogates close to or worse than the `Mean baseline` on ρ and T, and causes unregularized input concatenation (`Concat-MLP`) to diverge on `VHVC/large` ($\text{nMAE} > 10^3$) whereas multiplicative `FiLM-MLP` remains bounded (ρ: 5.29, T: 2.50, u: 0.787, v: 0.857).

4.4 Limitations

Three limitations of this study should be noted: (1) because the dataset is curated from legacy converged CFD-Post exports without original solver case/journal files or convergence histories, we cannot separate a physical flow-regime transition from convergence to an alternative steady numerical branch in the two near-stagnant cases `HHVC_P20_90_bar_0.05kgs_T_305` ($Q = 20$ W) and `HHVC_P30_90_bar_0.05kgs_T_305` ($Q = 30$ W), nor can we verify the specific k–ε sub-variant, wall-function treatment, or unexported hydrostatic reference density ρ_0; (2) plane-weighted mean CO_2 density $\bar{\rho}_{\text{plane},c}$ varies across steady-state cases at the

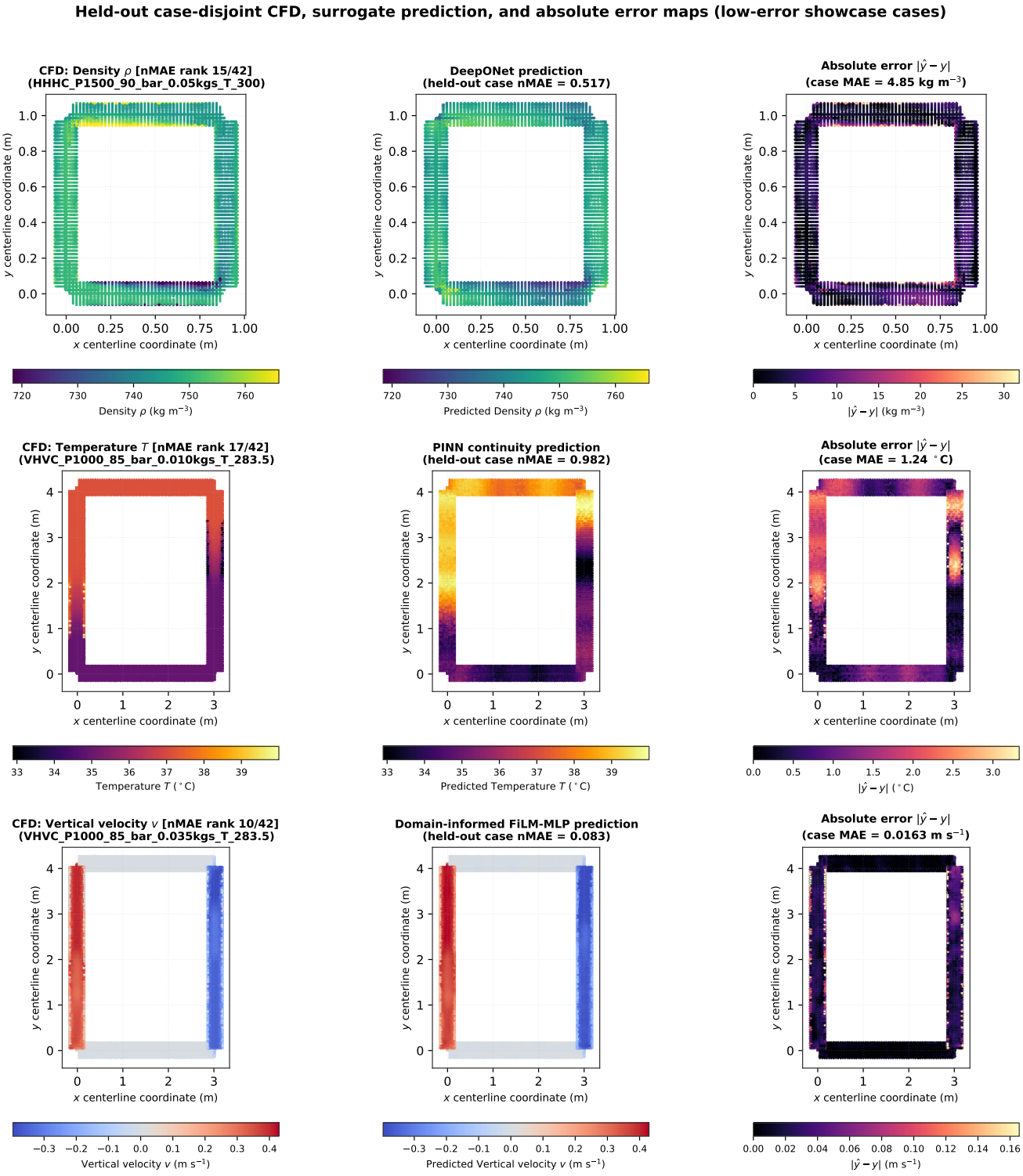


Figure 6: Showcase held-out out-of-fold spatial reconstructions (seed 2026) comparing reference CFD (left), surrogate prediction (middle), and absolute error $|\hat{y} - y|$ (right) for ρ (`DeepONet`, rank 15/42), T (`PINN continuity`, rank 17/42), and v (`Domain-informed FiLM-MLP`, rank 10/42).

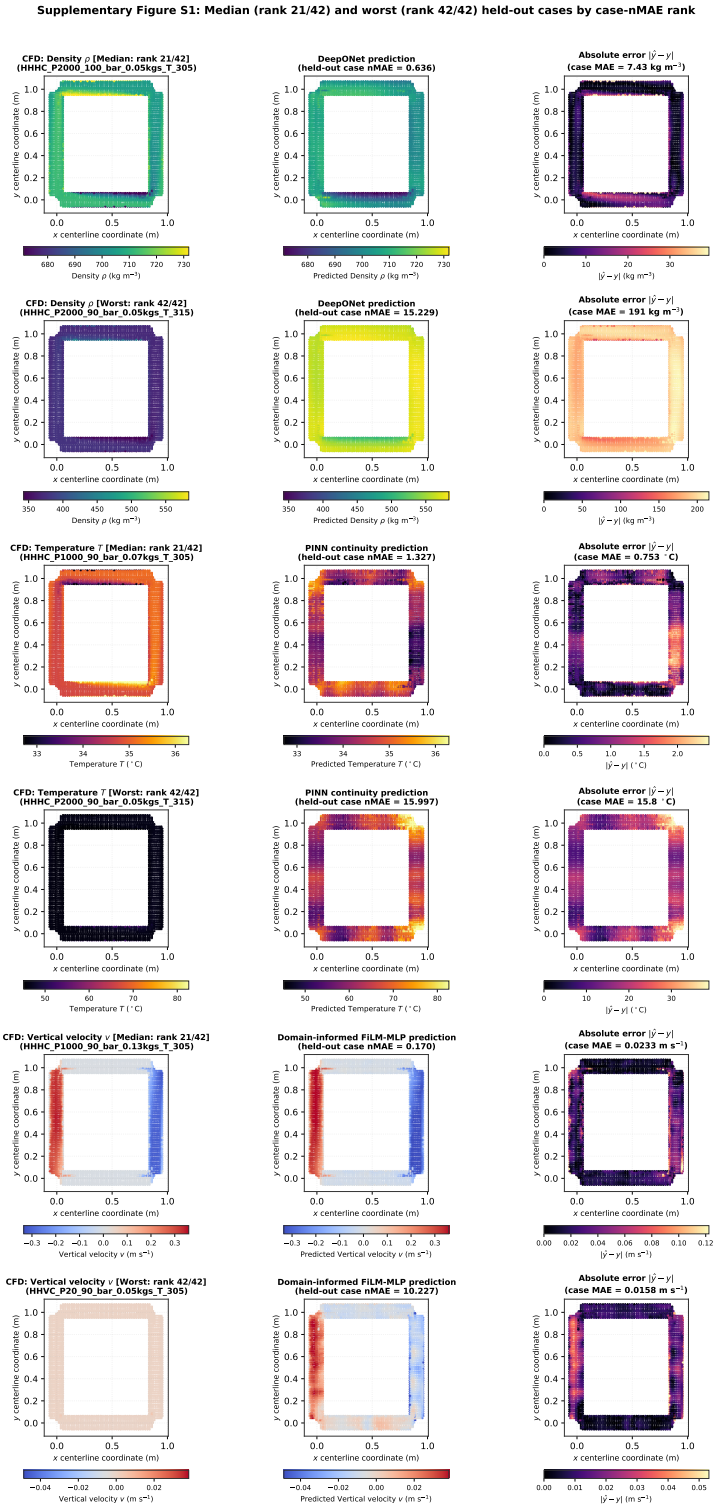


Figure 7: Median (rank 21/42, left block) and worst (rank 42/42, right block) held-out out-of-fold spatial reconstructions (seed `2026`) for ρ (`DeepONet`), T (`PINN continuity`), and v (`Domain-informed FiLM-MLP`). Rank-42/42 failures correspond to the sole $T_{\text{in}} = 315$ K case (`HHHC_P2000_90_bar_0.05kgs_T_315`) on ρ, T and the near-stagnant 20 W `HHVC` case on v.

Table 5: Leave-one-group-out structural/regime shift benchmark across the four groups (16 pooled models × 8 fields = 128 evaluations per group). Top rows report the three-seed mean median case nMAE on $(\rho, T, \varepsilon, k, v)$ for a fixed preselected seven-model shortlist; the bottom block lists the retrospective best pooled model per field (p_{gauge} via level MAE in kPa and $\text{nRMSE}_{\text{res}}$).

Block / Model	Field(s)	HHHC/small ($n = 20$)	HHVC/small ($n = 5$)	VHVC/small ($n = 4$)	VHVC/large ($n = 13$)
Fixed preselected shortlist: median case nMAE tuple (ρ / T / ε / k / v)					
Mean baseline	$\rho/T/\varepsilon/k/v$	5.11/9.57/5.85/4.63/0.81	13.3/21.8/15.7/9.95/0.95	4.52/8.51/9.04/6.79/1.08	5.57/1.70/0.43/1.89/0.84
Extra Trees	$\rho/T/\varepsilon/k/v$	7.00/11.6/0.41/1.03/1.01	1.17/3.34/0.33/0.62/1.21	2.25/3.74/0.37/0.99/0.89	4.36/1.65/0.44/1.98/0.77
XGBoost	$\rho/T/\varepsilon/k/v$	11.9/18.1/7.51/10.7/1.19	1.39/3.05/0.31/0.52/1.41	**0.98**/**1.61**/0.28/0.47/**0.69**	5.85/1.73/7.22/1.83/0.79
DeepONet	$\rho/T/\varepsilon/k/v$	9.66/68.8/0.41/0.97/0.99	6.11/7.94/0.39/0.70/1.49	3.50/7.18/0.37/0.85/0.78	5.90/2.51/ $>10^3$/ $>10^3$/0.82
PINN continuity	$\rho/T/\varepsilon/k/v$	5.48/44.2/1.01/1.58/0.73	4.94/19.2/0.49/0.66/1.11	6.57/12.5/18.9/2.13/0.77	5.36/3.18/4.46/2.47/0.91
Concat-MLP	$\rho/T/\varepsilon/k/v$	12.9/130/1.29/1.31/0.88	17.1/34.7/29.5/1.60/1.53	3.24/11.6/ $>10^3$/14.0/2.53	$>10^3$/ $>10^3$/ $>10^3$/ $>10^3$/ $>10^3$
FiLM-MLP	$\rho/T/\varepsilon/k/v$	12.1/113/6.82/1.30/1.05	12.1/13.9/0.90/0.88/1.14	7.42/16.4/14.4/1.09/1.22	5.29/2.50/25.0/1.87/0.86
Retrospective best pooled model across all 16 candidates (Supplementary Table S9)					
Best on ρ / T	nMAE	GNN (4.60) / Mean (9.57)	ET (1.17) / XGB (3.05)	XGB (0.98) / XGB (1.61)	RF (3.85) / ET (1.65)
Best on ε / k	nMAE	F-FNO (0.40) / F-FNO (0.65)	RF (0.28) / Ridge (0.44)	XGB (0.28) / XGB (0.47)	RF (0.42) / PINN-d (1.16)
Best on u / v / w	nMAE	PINN-c (0.69) / PINN-d (0.72) / Mean (0.38)	CB (0.79) / CB (0.67) / Mean (0.38)	XGB (1.12) / XGB (0.69) / Mean (0.39)	XGB (0.59) / ET (0.77) / Mean (0.23)
Best on p_{gauge}	Level / $\text{nRMSE}_{\text{res}}$	GNN (123) / Mean (0.196)	XGB (557) / Mean (0.186)	Mean (128) / Mean (0.185)	LNO (248) / XGB (0.988)

same $(p_{\text{op}}, T_{\text{in}})$ (Section 2.2), so the cases represent independent steady operating points rather than a strictly constant-inventory trajectory; and (3) seeds 2026, 2027, and 2028 re-randomize the five-fold stratified case splits in addition to model initialization, and the small group sizes in HHVC/small ($n = 5$) and VHVC/small ($n = 4$) limit statistical power under group holdout.

5 Conclusions

We have presented a case-disjoint surrogate benchmark across 42 steady-state sCO_2 natural-circulation loop CFD simulations (5,451,341 primary sCO_2 nodes). Our main conclusions are:

1. **Case-disjoint validation and two-part pressure evaluation are essential:** Across the three strictly matched-configuration tree baselines (RF, ET, CB), 80/20 random node splits understate five-fold case-disjoint case-macro MAE by 6.78 × –8.25× on density, 11.29 × –30.08× on gauge pressure, and 5.62 × –10.24× on temperature (6.78 × –10.37× and 11.29 × –87.54× across all four tree baselines including XGBoost), while Mean and Ridge controls remain near 0.95 × –1.26×. Separating gauge pressure into its case-level mean offset and centered spatial residual—referenced against the zero-residual baseline ($\text{RMSE}_{\text{res}} = 2.89$ kPa, $\text{nRMSE}_{\text{res}} = 0.202$)—prevents between-case pressurization shifts from obscuring within-loop gradient accuracy.

2. **Different physical fields favor distinct inductive biases:** Under three-seed five-fold case-disjoint cross-validation, multiplicative coordinate modulation (Domain-informed FiLM-MLP) leads all pooled models across all three velocity components (nMAE = 0.162, 0.175, 0.272) and both turbulence fields (nMAE = 0.192 on ε, 0.242 on k); DeepONet (nMAE = 0.837) and PINN continuity (nMAE = 1.350) lead among pooled models on density and temperature; and within-group POD-RBF achieves the lowest within-case nMAE on temperature (0.974), ε (0.177), and k (0.237) when interpolating on a fixed mesh, whereas uniform voxelized spectral operators (F-FNO, U-FNO) are bottlenecked by the 64 × 64 × 16 voxelization floor (Lattice floor (oracle), nMAE = 0.196–0.415) on slender pipe geometries ($D_i/L < 1.7\%$).

3. **Cross-group generalization requires multi-scale coverage:** Leave-one-group-out tests (128 model–field evaluations per group) show that pooled tree and neural models beat the Mean baseline when the loop scale and orientation are covered in training (VHVC/small), but degrade close to or worse than the Mean baseline under combined 3.4 × –4.1× geometric and operating-regime shifts (VHVC/large, HHHC/small).

Data and Code Availability

All preprocessing, benchmark training, statistical testing, and plotting scripts, along with the three-seed metric tables, pressure/density audit CSV, and supplementary material (Section S1, Figure S1, and Tables S0a–S10), are publicly available at https://github.com/Barghav777/SCWR_3D.

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